Power Sprayer



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 6 labs

Covers these lab items: Power sprayer

What it is used for

A power sprayer uses a motor or engine-driven pump to apply liquid over crops, plants, or approved surfaces. Correct chemical handling, pressure control, and maintenance help prevent exposure, leaks, and equipment damage.

Main parts

- Motor or engine drives the spray pump.
- Tank stores the spray liquid on tank-equipped models.
- Pump creates pressure to move liquid through the system.
- Pressure gauge displays operating pressure where fitted.
- Hose carries liquid from the pump to the spray lance.
- Lance and nozzle direct the spray toward the target.
- Pressure regulator and control valve adjust output on applicable models.

Before you start

1. Read the product label and sprayer manual before preparing the spray mixture.
2. Use only approved substances and follow the specified dilution and application rate.
3. Wear the required PPE, including chemical-resistant gloves, protective clothing, footwear, and eye protection.
4. Inspect the pump, hoses, tank, seals, clamps, lance, nozzle, and power source for damage.
5. Check that the suction line and filters are clean and all fittings are secure.
6. Choose a ventilated mixing area away from people, animals, food, and water sources.

How to use

1. Prepare the mixture according to the product label and fill the tank without overfilling.
2. Secure the tank cap and check that all hose connections are tight.
3. Position the sprayer on stable ground and ensure the intake has an adequate liquid supply.
4. Start the motor or engine according to the manual and check for leaks.
5. Adjust pressure to the recommended setting for the nozzle and task.
6. Test the spray pattern in a designated safe area before treating the target.
7. Hold the lance securely and apply the specified amount evenly over the target area.
8. Monitor pressure, flow, and hose condition during operation.
9. Stop the machine if leakage, blockage, abnormal vibration, or chemical exposure occurs.
10. Switch off, release pressure safely, clean the equipment as directed, and wash hands after removing PPE.

⚠ Safety

- Never operate the sprayer without the PPE specified on the product label.
- Do not point the spray at people, animals, food, or drinking-water sources.
- Never open pressurised fittings or clear a nozzle while the pump is running.
- Keep the engine exhaust away from enclosed areas and ignition sources.
- Do not mix incompatible chemicals or use unlabelled substances.
- Dispose of leftover mixture and wash water according to the product label and local requirements.

Care & storage

- Release pressure and isolate the power source before cleaning or maintenance.
- Rinse the tank, pump, hose, and nozzle using the approved cleaning procedure.
- Inspect seals, filters, hoses, clamps, and nozzle openings regularly for wear or blockage.
- Store the clean equipment in a secure, dry area and arrange servicing as instructed.

Common mistakes to avoid

- Running the pump without an adequate liquid supply when dry running is not permitted.
- Setting pressure beyond the rated limit of the hose or equipment.
- Opening pressurised connections or clearing a blockage with the pump running.
- Spraying without PPE or disposing of chemical waste improperly.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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